

From Queries to Qubits: Data Management Meets Quantum Computing

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Abstract

Over the past several years, quantum computing has turned from a purely theoretical concept into a technology that is widely available via cloud offerings from multiple providers. Several manufacturers are now producing noisy intermediate-scale quantum hardware whose capabilities are growing at a fast pace. These developments have inspired a growing number of publications in the database community, as well as the creation of several workshops dedicated to exploring topics at the intersection of data management and quantum computing. This panel brings together experts from the quantum computing and data management communities, working in industry and academia, to discuss challenges and opportunities in this emerging research field.

CCS Concepts

• Information systems → Data management systems; • Hardware → Quantum computation.

Keywords

quantum computing, data management

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1 Panel Overview

Several scientific breakthroughs over the last decade have turned quantum computing [1–3] from merely a theoretical concept to a technology that is starting to see adoption in practice. Various manufacturers, ranging from the likes of Google, IBM, and Intel to more specialized companies such as D-Wave, IonQ, and Rigetti, have started to produce early-stage hardware that implement Noisy Intermediate-Scale Quantum computing (NISQ) exploiting a multitude of approaches. Inspired by these advances, other providers including Fuhitsu, NEC, and Toshiba, have started to offer classical accelerators mirroring the inner workings of actual quantum hardware to gain performance advantages for certain types of problems. Many of these devices are nowadays available to broad shares of the population via cloud services like Amazon Braket or D-Wave's

Leap, making it easy to experiment without purchasing expensive hardware upfront. At the same time, the software ecosystem for quantum hardware is evolving quickly, facilitating their use for novel problems.

These technological advances raise two questions from a data management perspective. First, how can we leverage quantum accelerators and related technologies to solve problems relevant to the database community? And, second, how can the database community support advances in the domain of quantum computing?

This panel brings together a diverse mix of experts, covering industry as well as academia, and covering the quantum computing as well as the data management communities, to explore these questions, to give the audience an impression of the current state, and to discuss challenges and opportunities in this emerging field. The panel will start with short position statements by the panelists, followed by discussions, centering on pre-selected questions and questions by the audience.

2 Panel Participants

Floris Geerts. Floris Geerts is a professor at the University of Antwerp. He received his M.Sc. in Mathematics from the University of Ghent (Belgium) in 1995, and a Ph.D. in Computer Science from the University of Hasselt (Belgium) in 2001. He has been postdoctoral researcher at the University of Helsinki (Finland) and was Senior Research Fellow at the University of Edinburgh (UK) for seven years. He joined the University of Antwerp as Research Professor in 2011. His research interests include principles of data management, data quality, graph neural networks, and topics at the intersection between quantum computing and data management, covering the use of data management ideas in the context of quantum computing (e.g., for quantum simulation), as well as the use of quantum technologies to improve database systems (e.g., for improved privacy).

Rihan Hai. Rihan Hai is an assistant professor at the Web Information Systems (WIS) group at TU Delft and the leader of the InfiniData Team. She approaches quantum data management from two directions: classical simulation of quantum computation, using data management techniques, and the prototyping of quantum data systems. Her goal is to drive research in the current NISQ era of quantum computing while building a solid foundation for future applications of quantum computing in data management and AI. Before joining TU Delft, Rihan received her Ph.D. degree “summa cum laude” from RWTH Aachen University, Germany, and her BEng and MEng from Tsinghua University, China.

Manish Kesarwani. Manish Kesarwani is an Advisory Research Scientist at IBM Research, India. With over a decade of research experience at IBM Research, he has led projects at the intersection



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of RDBMS, Quantum Computing, AI and Applied Cryptography. He holds more than 40 patents, has served on program and organizing committees of top-tier conferences, and was recognized among IBM's Top Global Contributors and Top Inventors in 2025. His recent work focuses on fundamental challenges in hosting RDBMS components on quantum platforms.

Wolfgang Maurer. Wolfgang Maurer is a Professor of Quantum Computer Science at the Technical University of Applied Sciences Regensburg, a R&D professor in the Munich Quantum Valley, and a Senior Research Scientist at Siemens AG, Foundational Technologies, Munich. His interests focus on software and systems engineering for quantum computing and other future compute technologies. He obtained his PhD at the Max Planck Institute for the Science of Light, and has published strongly multi-disciplinary work in venues and journals from Nature Photonics and Physical Review Letters via ICSE and TSE to SIGMOD and VLDB.

Krishnakumar Sabapathy Krishnakumar Sabapathy, head of research in Quantum at Fujitsu, India, is a quantum information scientist with over a decade of post-PhD research experience across academia and industry in India, Europe, and Canada. He specializes in building practical R&D solutions across the quantum technology stack, spanning quantum theory, algorithms, applications, and software. Currently focused on enabling enterprise and research users to harness quantum hardware effectively, Krishnakumar also contributes to advancing hardware development through deep, application-driven insights.

Ibrahim Sabek. Ibrahim Sabek is an Assistant Professor of Computer Science at the University of Southern California. He leads the Next-generation Data-Intensive Systems Group (NexDIG), focused

on building the next generation of data management, processing, and analysis systems using machine learning and quantum computing. Before joining USC, Ibrahim was a Postdoctoral Associate in the Data Systems Group at MIT CSAIL and an NSF/CRA Computing Innovation Fellow (2020-2023). He completed his PhD in Computer Science from the University of Minnesota, Twin Cities, in January 2020. He was awarded the 2025 Google ML and Systems Junior Faculty Award, the 2024 and 2025 Google Data Analytics and Insights (DANI) Awards, and the University of Minnesota's Best Doctoral Dissertation Honorable Mention Award in 2021 for his PhD work, among others.

Kurt Stockinger. Kurt Stockinger is a Professor of Computer Science at the Zurich University of Applied Sciences and the University of Zurich, Switzerland. He has held research positions at the University of Washington, CERN, Berkeley National Laboratory, and the California Institute of Technology. His work spans areas such as natural language query processing, machine learning, and quantum computing. In the latter area, his publications apply quantum computing to problems such as query optimization, cardinality estimation, entity matching, and anomaly detection. He holds a Ph.D. in computer science from the University of Vienna, Austria, under the supervision of CERN, Switzerland.

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